

Generative AI for All and for Humanity: From Zero to Hero with Open Data, Space Computing, and Sustainable Smart Cities

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Abstract

Generative artificial intelligence is reshaping creativity, learning, and problem solving—a renaissance for all and for humanity. This one-hour workshop guides educators and learners “from zero to hero” by linking generative AI with open data, space computing, and sustainable smart-city applications. Participants will curate and publish 3D urban datasets with clear documentation, transform them into interactive models, and simulate energy-aware scenarios such as drone routing. The workshop emphasizes responsible data governance and ethical practice, highlighting principles that call for data to be easy to find and reuse as well as respectful of community rights and benefits. Outcomes are aligned with the United Nations Sustainable Development Goals to connect technical practice with global challenges. Participants will leave with openly licensed teaching resources, including a rubric, workflow, and project templates that integrate responsible AI into curricula and incubate capstone proposals in computer graphics and interactive technologies.

CCS Concepts

• **Computing methodologies** → **Computer graphics**; • **Applied computing** → **Collaborative learning**; **Interactive learning environments**; • **Information systems** → **Collaborative and social computing systems and tools**; • **Social and professional topics** → **Computing education**.

Keywords

Generative AI, Open Data, Space Computing, Smart Cities, Education, FAIR Principles, CARE Principles, SDGs, Drones, AR/VR, Blockchain Platforms, Web3, Responsible AI

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1 Overview

The SIGGRAPH Asia 2025 Educator’s Forum emphasizes practical, hands-on learning that participants can transfer to classrooms in computer graphics, interactive technologies, animation, and digital art. Our workshop responds by framing generative AI through the AI triad—data, algorithms, and compute power—and by demonstrating how principled governance of each pillar enables equitable access and reproducible outcomes in graphics and interactivity, addressing the challenges of responsible AI in the generative era [Kenthapadi et al. 2023]. In this triad, data provide the substrate for learning; algorithms transform data into expressive capabilities across imaging, geometry, and interaction; and computing power scales these capabilities from classroom prototypes to research-grade pipelines. Neglecting any pillar risks “garbage in, garbage out,” reproducibility failures, or inequitable access. Conversely, balancing the triad empowers educators and learners to harness generative AI in building open, verifiable, and societally relevant experiences.

In concrete terms, the workshop equips participants to curate and govern datasets using interoperable metadata; to connect those datasets to algorithms through programmatic, reusable loaders; and to transform them into 3D models and interactive simulations that animate space-computing and smart-city scenarios. We introduce the FAIR Principles for scientific data stewardship to scaffold discoverability and interoperability [Wilkinson et al. 2016], and we complement them with the CARE Principles to center collective benefit, authority to control, responsibility, and ethics [Jennings et al. 2023]. Learning outcomes are aligned with the United Nations Sustainable Development Goals to situate creative work within globally recognized challenges [UNS 2025]. These foundations are increasingly salient in domains such as the low-altitude drone economy, where safety and infrastructure shape deployment [Wang and Ran 2025], and in blockchain platforms and the Web3 ecosystem,

where governance choices influence inclusion [Chemaya et al. 2025; Liu et al. 2022; Xiao et al. 2024; Yan et al. 2025].

2 Literature Review

Our workshop builds upon, and advances, three strands of work represented in the SIGGRAPH Asia Educator’s Forum.

2.1 Sustainability in Education Pipelines

Prior contributions emphasized sustainable collaboration between production and academia, environmental and curricular challenges in VFX education, and modernization of CS guidelines [Helzle et al. 2024; Matsunaga et al. 2023; Reiser and Lait 2024]. We extend this agenda upstream to *data governance*. By embedding FAIR and CARE into dataset design and reuse, and aligning outcomes with the SDGs, we target sustainability at the source: how data are collected, documented, shared, and critiqued before downstream rendering, animation, or XR integration.

2.2 3D Computer Graphics and Interactive Pedagogy

Educators have advanced 3D look development via reverse engineering and mind mapping, immersive learning of vector mathematics in VR, and accessible introductions to WebGL [Angel and Shreiner 2020; Cabanag 2023; Matsunaga and Miyata 2024]. We build on these methods by adding a reproducible, metadata-driven data layer beneath graphics tasks. Activity A creates small but teachable open datasets documented with Croissant metadata for ML-ready use [Akhtar et al. 2024]. Activity B converts these datasets into 3D assets and interactive simulations, closing the loop between data curation, model training or inference, and visualization.

2.3 Open Access and Interactive Techniques

Recent work championed open-source frameworks for graphics learning and reflective assessment practices in game programming [Hooper et al. 2024; Petropoulos et al. 2024]. Our contribution complements these by standardizing the *data interface* that fuels such frameworks and by providing an assessment rubric that explicitly evaluates documentation, reusability, prototype quality, and ethical framing. This enables instructors to integrate open tools while maintaining principled governance of the data that power them.

3 Workshop Description

The agenda guides participants from open-data governance to applied modeling in space computing and sustainable smart cities. The opening introduces the AI triad and shows how FAIR and CARE principles operationalize responsible practice in computer graphics and interactive technologies. Participants then curate urban 3D scans (e.g., photogrammetry or LiDAR), document them with Croissant metadata, and publish on Hugging Face using the datasets library [Hug 2025; Akhtar et al. 2024]. Emphasis is placed on documentation standards that ensure reusability and scientific impact, with the potential for later scaling through initiatives such as the AWS Registry of Open Data [AWS 2025].

In the next phase, participants transform the published datasets into 3D models and web-based simulations (e.g., shader pipelines or

Table 1: Agenda: Stages connecting open data publication, space-computing pipelines, and sustainable smart-city simulation.

00–05 min ☞ Opening	Goals and Orientation — Introduce the workshop goals, highlight the AI triad (data–algorithms–compute), and connect these to computer graphics, interactive techniques, and the Sustainable Development Goals (SDGs).
05–15 min ➤ Framing	Principles for Data Governance — Present the FAIR and CARE frameworks, emphasizing how open datasets published on Hugging Face can maximize reuse, ensure transparency, and scale toward scientific impact while engaging with Web3 governance considerations [UNS 2025; Jennings et al. 2023; Wilkinson et al. 2016].
15–30 min ♦ Activity A	Open Data Curation — Guide participants in curating 3D-scan-derived urban data, documenting them with Croissant metadata, and publishing them on Hugging Face with ready-to-use datasets loaders. Discuss quality criteria and scaling pathways to repositories such as the AWS Registry of Open Data [AWS 2025; Hug 2025; Akhtar et al. 2024].
30–45 min ★ Activity B	Space Computing and Smart Cities — Transform the curated datasets into 3D models and implement web-based interactive simulations (WebGL-style). Use the models to design energy-efficient drone-routing strategies for sustainable logistics and climate action [Angel and Shreiner 2020; Wang and Ran 2025].
45–55 min ◆ Discussion	Critical Reflection — Engage participants in analyzing how governance decisions influence modeling outcomes. Discuss blockchain-platform and Web3 trade-offs, reproducibility challenges, and connect insights to prior SIGGRAPH Asia education work on sustainability, 3D graphics, and open-source techniques [Cabanag 2023; Hooper et al. 2024; Matsunaga and Miyata 2024; Petropoulos et al. 2024].
55–60 min ☞ Incubation	Capstone Co-Design — Conclude by co-developing capstone project proposals that integrate open datasets, 3D modeling, and space-computing pipelines to prototype sustainable smart-city applications such as drone corridor planning, energy-aware routing, and SDG-aligned urban visualization.

Legend: ☞ Opening ➤ Lecture ♦ Hands-on ★ Build ◆ Discussion ☞ Incubation.

WebGL rendering [Angel and Shreiner 2020]) to explore smart-city and space-computing scenarios. They prototype metaverse-ready assets and low-altitude economy use cases such as energy-efficient drone routing, linking technical outputs to climate action and the SDGs.

3.1 Agenda

Table 1 aligns each phase with the workshop themes of open data, space computing, and sustainable smart cities. Open data emphasizes responsible publication and governance, space computing highlights data-to-model pipelines for navigation and simulation, and sustainable smart cities focus on interactive planning for low-altitude scenarios and SDG-linked climate action. The final segment incubates capstone proposals that extend the workshop outputs into broader course or program initiatives.

3.2 Speaker Biographies

Collectively, the speakers span cloud platforms and AI infrastructure, medical imaging and reliable ML, climate resilience and sustainability engineering, multimedia quality and perception, and socio-technical economics. Their combined expertise offers a coherent, practice-grounded perspective on empowering learners “from zero to hero” by balancing the AI triad and embedding principled

data governance into graphics and interactive-technology education.

Dongping Liu is Senior Industry Business Development Manager for Higher Education and Research at Amazon Web Services. He received his Ph.D. in Physics from the Chinese Academy of Sciences and completed postdoctoral research at McGill University, subsequently serving as Associate Researcher at the Chinese Academy of Sciences. His expertise spans AI, deep learning, and high-performance computing with an emphasis on educational digital transformation. A recipient of the Beijing Science & Technology Award, he has authored more than thirty publications and holds over ten patents. He currently serves as Chairman of the Hong Kong branch of the China Computer Federation Young Computer Scientist and Engineer Forum.¹

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Shiqi Wang is Associate Dean of the College of Computing and Professor in the Department of Computer Science at the City University of Hong Kong. An internationally recognized scholar in multimedia AI, perceptual quality assessment, and image and video analysis, he has produced more than 400 publications with an h-index of 64 and received the IEEE Multimedia Rising Star Award as well as multiple Best Paper Awards. His research advances perceptual metrics and optimization for AI-generated and natural media.⁴

Luyao Zhang is Assistant Professor of Economics and Senior Research Scientist at the Digital Technology Innovation Center at Duke Kunshan University, where she co-leads the Contemporary China Digital Technology and Society Cluster. Her scholarship bridges generative AI, interactive techniques, and social science, with publications in CSCW, NeurIPS, ACM CCS, ACM S&P, and *Scientific Data*. She pioneers project-based pedagogy that links open data, blockchain platforms, the Web3 ecosystem, and graphics to the SDGs through experiential learning with industry and community partners.⁵

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3.3 Materials and Requirements

To facilitate immediate adoption, the workshop will provide a collection of openly licensed resources, hosted in a public GitHub repository. Table 2 summarizes the materials and technical requirements.

Table 2: Workshop Materials and Technical Requirements. All resources will be openly licensed and provided in reusable formats to support integration into diverse educational contexts.

Slides & Notes	An instructor slide deck introducing the AI triad, FAIR/CARE governance principles, and SDG framing. Annotated notes will be provided to allow educators to adapt the material directly into their own courses.
Activity Guides	Step-by-step written guides for Activity A (dataset curation and publication) and Activity B (3D modeling and interactive simulation), designed to scaffold both novice and advanced participants.
Metadata Templates	Croissant JSON-LD exemplars for documenting 3D-scan datasets such as building façades, corridors, and rooftops. Templates emphasize interoperability, discoverability, and licensing clarity.
Jupyter Notebooks	Interactive Python notebooks demonstrating (1) dataset publication on Hugging Face, (2) programmatic loading with the datasets library, and (3) energy-aware drone path-planning built upon the published data.
Assessment Tools	Rubrics and checklists aligned with Table 3, enabling instructors to evaluate dataset documentation quality, reusability, and SDG alignment in a transparent and reproducible way.
Technical Requirements	<i>Primary:</i> Google Colab (free tier), supporting Python 3.10+ and required libraries without local setup. <i>Alternative:</i> A laptop with Python 3.10+, JupyterLab or Notebook, and internet access, with packages datasets, numpy, pandas, matplotlib, and networkx. <i>Optional:</i> iOS device with Reality Composer or a WebGL-enabled browser for visualization; AWS free-tier account for testing dataset hosting.

3.4 Assessment and Rubric

Evaluation emphasizes demonstrable learning aligned to responsible data practice, reproducibility, and human-centered design for sustainability. Table 3 formalizes expectations across four dimensions: the completeness and clarity of dataset documentation; the reusability of the dataset through programmatic loading and open publication; the communicative power of the interactive 3D simulation; and the explicit alignment to FAIR, CARE, and the SDGs, including energy- or carbon-aware reasoning.

3.5 Illustrative Workflow

Figure 1 summarizes the process-first sequence and shows how the agenda in Table 1 maps onto the assessment criteria in Table 3. By articulating transitions from governance framing to open publication and from modeling to interactive simulation and capstone incubation, the diagram helps students document process rather than present only polished artifacts.

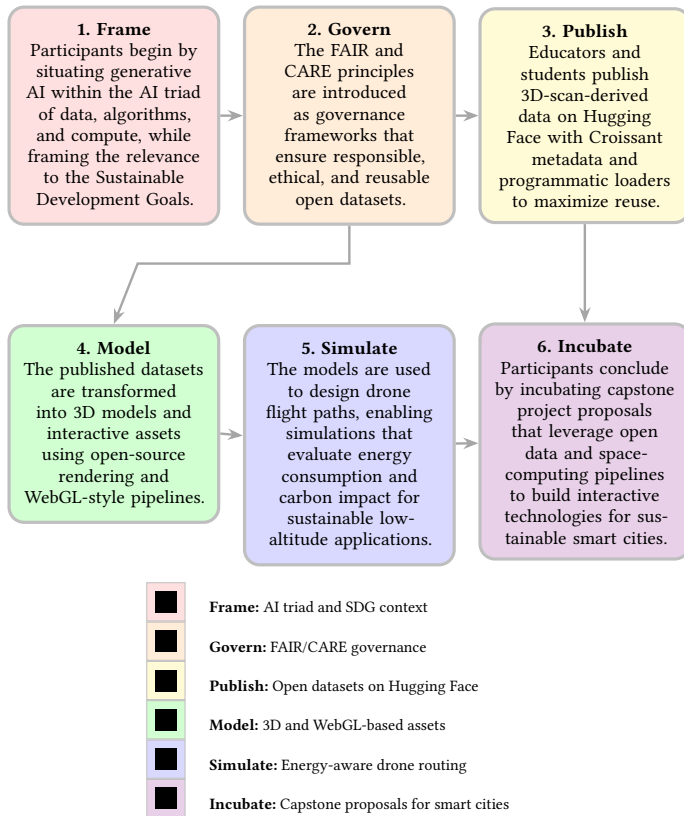
3.6 Expected Outcomes

Participants will publish a compact, well-documented open dataset with Croissant metadata and a datasets loader on Hugging Face; will convert those data into 3D assets; and will implement a web-based interactive simulation that explores energy-efficient drone routing for smart-city or space-computing use cases. They will leave with a rubric and workflow adaptable to varied curricula and a principled rationale for aligning generative AI practice with the AI

Table 3: Assessment rubric with explicit FAIR/CARE and SDG integration. Shaded headers indicate low, medium, and high levels of evidence; icons highlight performance targets.

Dimension	Low	Medium	High
Data Documentation	✗ Provides only a minimal schema with little explanation.	● Includes partial metadata but lacks consistency or clarity.	✓ Supplies complete Croissant metadata with a clear license and comprehensive README.
Reusability & Publication	✗ Dataset cannot be loaded and remains private.	● Dataset loads locally but has limited documentation.	✓ Dataset is openly published with a Hugging Face loader, version control, and clear instructions for reuse.
Interactive 3D Simulation	✗ Prototype is incomplete or static with little interactivity.	● Prototype provides basic navigation and limited visualization.	✓ Prototype is a fully web-based, interpretable simulation tied to the dataset, supporting path and energy optimization.
Responsible Framing & SDGs	✗ Work shows no awareness of ethics or societal impact.	● Work acknowledges potential risks and benefits in general terms.	✓ Work explicitly integrates FAIR and CARE principles, aligns with SDGs, and demonstrates energy-conscious reasoning.

Process: From Open Data to Space Computing and Sustainable Smart Cities



Mapping: The stages mirror Table 1 and are evaluated through Table 3.

Figure 1: Two-row workflow from framing and governance to open publication, modeling, simulation, and incubation of capstone projects. The rainbow palette highlights progression and continuity across phases.

triad and with FAIR, CARE, and the SDGs. The final discussion will yield incubated capstone project proposals that extend workshop

artifacts into deployable, research-backed educational products for sustainable smart cities within blockchain platforms and the Web3 ecosystem.

4 Conclusion: Connection to the Generative Renaissance

This workshop advances the SIGGRAPH Asia 2025 theme of the Generative Renaissance by demonstrating how data, algorithms, and compute power can be responsibly orchestrated for human-centered learning in computer graphics and interactive techniques. By grounding practical activities in the FAIR and CARE principles and the SDGs, it fosters intrinsic motivation, critical reflection, and civic purpose while equipping educators with ready-to-use frameworks that promote sustainability, reproducibility, and equity.

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